

The dark side of diversification: The case of US financial holding companies

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Abstract

Potential diversification benefits are one reason why US financial holding companies are offering a growing range of financial services. This paper examines whether the observed shift toward activities that generate fees, trading revenue, and other non-interest income has improved the performance of US financial holding companies (FHCs) from 1997 to 2002. We find evidence that diversification benefits exist between FHCs, but these gains are offset by the increased exposure to non-interest activities, which are much more volatile but not necessarily more profitable than interest-generating activities. Within FHCs, however, marginal increases in revenue diversification are not associated with better performance, while marginal increases in non-interest income are still associated with lower risk-adjusted profits. The key finding that diversification gains are more than offset by the costs of increased exposure to volatile activities represents the dark side of the search for diversification benefits and has implications for supervisors, managers, investors, and borrowers.

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1. Introduction

The Gramm–Leach–Bliley Act of 1999 (GLBA) capped a decade of fundamental change for the US banking industry by removing many restrictions on financial service providers.¹ While US banks have been shifting away from lending activities and towards broader financial services for decades, GLBA opened the way for full financial integration by explicitly allowing financial holding companies (FHCs) and their subsidiaries to engage in a host of new activities like brokerage, advisory, and underwriting. Most large banking institutions, for example, now routinely offer these services.²

This paper asks a straightforward question about this shift toward activities that generate non-interest income: do more diversified financial holding companies (FHCs) systematically outperform more concentrated ones? To identify diversified FHCs, we turn to the income statement and focus on the sources of net operating revenue.³ An FHC that earns all of its revenue from net interest income, for example, is considered concentrated, while an FHC that relies evenly on net interest income and non-interest income is considered diversified. We compare this measure of revenue diversification to risk-adjusted profits to gauge the impact of the shift away from net interest income.

Several recent papers have examined a similar question for US commercial banks and have typically found little evidence of gains from the shift toward fee and other non-interest income.⁴ This paper makes several contributions regarding both methodology and data that help us to better understand this conclusion. On the methodology side, we introduce a new framework for decomposing the impact of strategic shifts into a “direct exposure effect” (a greater reliance on new activities) and an “indirect diversification effect” (the resultant change in revenue concentration). This allows us to identify the channels through which these shifts impact performance and thereby isolate any diversification gains. We also utilize panel data to perform a “within” analysis and quantify the impact of changes in diversification for a given FHC. This removes an important omitted variable concern found in earlier studies that performed purely cross-sectional analyses and proves to be quite important.

On the data side, we examine the consolidated financial holding company (FHC), rather than the subsidiary commercial bank. Managers presumably make decisions with the goals of the entire institution in mind and regulators monitor the health of the entire institution, so this focus has the advantages of analyzing the more appro-

¹ See Furlong (2000) and Fay (2000) for overviews of GLBA.

² Following GLBA, bank holding companies (BHCs) needed to convert to financial holding company (FHC) status to access many new powers. Most large commercial banks are now part of FHCs, and, for ease of exposition, we use the term FHC to refer to both bank and financial holding companies throughout the remainder of the paper.

³ Net operating revenue is a standard measure of bank revenue and equals the sum of net interest income and non-interest income.

⁴ Recent papers based on accounting data include DeYoung and Roland (2001) and Stiroh (2004a,b), while DeLong (2001) shows that diversifying bank mergers reduce value.

priate unit of observation and possibly uncovering diversification benefits between related entities. Finally, we examine a relatively recent and short period from 1997 to 2002 that was quite turbulent due to the Russian bond default and Long-Term Capital Management (LTCM)-induced liquidity crisis in 1998, the stock market boom and bust, and the slowdown in capital market activity. The short period avoids the difficulties with examining evolving institutions over longer periods of time, while the volatility is interesting because several FHCs explicitly stated that a more diversified strategy helped to stabilize revenue over this period. Moreover, we can compare and contrast these results to earlier studies to build a more compelling case for the generality of the conclusions.

Our results begin with a cross-sectional analysis of over 1800 FHCs that operated between 1997:Q1 and 2002:Q4. Across FHCs, we find that revenue diversification is associated with higher risk-adjusted profits, but these gains are typically offset by the costs of increased exposure to volatile non-interest activities. In a portfolio framework, this is analogous to benefits from a covariance effect, but costs from increased exposure to a relatively volatile investment. Thus, diversification benefits seem to exist, but they are obscured by the direct exposure to high-volatility activities like trading. We note that our focus is on unconditional risk, measured with accounting data as the volatility of profits, because this is a relevant measure for supervisors, investors, managers, and borrowers and allows us to include a broad sample of FHCs of different sizes.

Consider the median FHC, which earned about 0.18 of net operating revenue from non-interest sources and had a risk-adjusted return on equity (mean return on equity divided by the standard deviation of the return on equity) of 4.0 from 1997 to 2002. For this FHC, a shift toward non-interest activities affects performance in two ways: a direct exposure effect from increased non-interest income and an indirect diversification effect as revenue becomes more diversified. Empirically, a one standard deviation increase in the non-interest share (about 0.10) is associated with a fall in the risk-adjusted return on equity of 0.06 due to the direct exposure effect, but an increase of 0.05 due to the diversification effect. The individual effects are substantial, but the net effect is insignificantly different from zero for the median FHC. For FHCs more concentrated in non-interest income (90th percentile non-interest share), however, a one standard deviation rise in the non-interest share is associated with a significant decline in risk-adjusted profits as the costs of the direct exposure to more volatile activities outweigh any diversification benefits. The higher volatility of non-interest income is the darkside of revenue diversification and has clear negative implications for supervisors, shareholders, managers, and borrowers, all of whom care about the mean and volatility of profits.

We then move beyond the cross-section and examine this relationship in a panel framework that links changes in strategic focus with changes in risk-adjusted performance. As in the cross-sectional analysis, increased reliance on non-interest income is strongly, negatively associated with risk-adjusted profits within FHCs. There is no evidence, however, that diversification affects performance within FHCs.

This difference in estimated diversification benefits in the cross-section and panel framework could reflect an omitted variable like management skill, scale, location,

or industry exposure that affects both diversification choices and FHC performance. That is, the decision to diversify is endogenous, so that a simple cross-sectional analysis may be biased if the diversification variable is correlated with other omitted factors that drive performance, e.g., if better managers both diversify and generate strong performance, then diversification itself may not be beneficial. In contrast, the negative impact of increased non-interest shares is robust to the inclusion of fixed effects, so an omitted variable explanation does not appear to be driving this key finding.

An alternative explanation is that expected and actual returns diverged due to unexpected economic shocks. For example, an FHC might expand operations and become more diversified if the expected returns compensate for the increased risk. An adverse economic shock that causes actual returns to fall below expected returns, however, would weaken the link between diversification and actual historical returns. Because these shocks impact the annual data in the panel analysis more than the six-year averages used in the cross-sectional analysis, it is plausible that short-term shocks explain the difference between panel and cross-sectional results. The consistency with earlier studies, however, suggests that this is not the whole story.

These results beg the question of why FHCs are moving steadily into these new activities. One explanation is that FHC managers may have overestimated the benefits of diversification. In the financial and industry press, for example, it is common to hear about the potential of “cross-selling” where a FHC sells multiple products to the same core customer base to reap economies of scope and gain diversification benefits. While this may open new revenue streams, they are more likely to be exposed to the same types of shocks, e.g., an industry slowdown or changing consumer preferences, so traditional diversification benefits would likely diminish. Alternatively, FHC managers may focus on expected profits and place relatively little weight on volatility. This might be reasonable if managers reap the gains of higher profits, but don’t bear all of the costs from increased risk, e.g., if managers are equity-holders, they have an incentive to take risk beyond what debt-holders and supervisors would prefer.

A third explanation for this shift are the non-profit maximizing motives discussed by Berger et al. (1999); Milbourn et al. (1999); Bliss and Rosen (2001); Houston et al. (2001), and Aggarwal and Samwick (2003), e.g., empire-building, over-diversification to protect firm-specific human capital, corporate control problems, or managerial hubris and self-interest. Finally, it could be just a short-run phenomenon due to adjustment costs associated with product-line expansion, an over-reaction to the lending problems of the late 1980s and early 1990s, or even simply bad luck. If true, risk-adjusted performance should improve as the necessary business practices, technology, scale are developed.

Whatever the explanation, recent acquisition and divestiture activity by US FHCs suggests that the worse performance may not be news to FHC managers. After a series of well-publicized acquisitions of investment banks, money management, and brokerage firms in the late 1990s, the industry has more recently seen large divestitures, e.g., US Bancorp’s spin-off of Piper Jaffray, FleetBoston’s closing of Robert-

son Stephens, and Citigroup's spin-off of Travelers Property Casualty.⁵ Similarly, the recent agreement between Bank of American and Fleet Boston, for example, was touted as a return to retail banking roots.⁶ One interpretation, albeit speculative, is that US bankers are now aware of the limitations of product-line expansion and are retrenching toward core competencies in order to avoid the products that adversely affect the bottom line.

2. Previous literature on performance and diversification

The question of whether diversified financial institutions outperform their more concentrated peers is a topic of active research. Saunders and Walters (1994); Reichert and Wall (2000); DeYoung and Roland (2001), and Stiroh (2004b) all provide detailed reviews of the literature, so this section focuses on the most relevant papers in order to highlight the contributions of the current study.

The empirical literature on financial firms has produced mixed evidence as to whether and how increased diversification affects performance.⁷ In an early survey, Saunders and Walters (1994) review 18 studies that examine whether non-bank activities reduce bank holding company (BHC) risk and indicate no consensus: 9 answer yes, 6 answer no, and 3 are mixed. These, and more recent studies, approach the risk question from a variety of perspectives: creation of synthetic or counterfactual mergers of banks with non-banks, analysis of actual operating results, and analysis of market reactions to diversification.

The most relevant comparisons for this study are the papers that examine the actual performance of banks with varying degrees of concentration and diversification. The general conclusion is that bank expansion into less traditional financial activities is associated with increased risk and lower returns. DeYoung and Roland (2001) find that a shift toward fee-based activities is associated with increased revenue volatility and a higher degree of total leverage, both of which imply greater earnings volatility for commercial banks. Stiroh (2004b) concludes that a greater reliance on non-interest income, particularly trading revenue, is associated with higher risk and lower risk-adjusted profits across commercial banks. A study of loan portfolio diversity by Acharya et al. (2002) reports that diversification of loans does not typically improve performance or reduce risk in Italian banks. Morgan and Samolyk (2003) examine geographic diversification and find similarly negative results: diversification is not associated with greater returns (ROE or ROA) or reduced risk. Finally, Stiroh

⁵ "A Painless Extraction: Closer Look at USB-Piper," Laura Mandaro, *American Banker*, February 21, 2003.

⁶ "Branching Out: Bank of America Bets on Consumer," Carrick Mollen and John Hechinger, *Wall Street Journal*, October 28, A1, 2003.

⁷ We note that classic finance theory suggests that firms should not expend resources on risk-reduction because an investor could diversify by buying a wide-range of securities. This issue is addressed at length in Section 3.

(2005) shows that increased exposure to non-interest income increases the volatility of equity market returns, but not the mean.

A few studies find some potential for gains from expansion into specific activities. Templeton and Severiens (1992) examine 54 bank holding companies from 1979 to 1986 and find that diversification (as measured by the share of market value not attributed to bank assets) is associated with lower variance of shareholder returns. Kwan (1998) examines the returns of banks' Section 20 subsidiaries and their commercial bank affiliates and finds that Section 20 subsidiaries are typically more risky and not necessarily more profitable than their commercial bank affiliates. Nonetheless, Kwan concludes that some diversification benefits may exist for commercial banks because of the low return correlation between securities and bank subsidiaries. Cornett et al. (2002) also report that evidence of gains from Section 20 subsidiaries as the industry-adjusted operating cash flow return on assets rises, while risks do not change significantly.

These studies paint a mixed picture about the performance effects of diversification in banking. This paper makes several contributions regarding both data and methodology. In terms of methodology, we present a new framework for decomposing the impact of strategic shifts into a direct exposure effect (a greater reliance on new activities) and an indirect diversification effect (the change in revenue concentration). This allows us to better understand the channels by which increased non-interest income and diversification impact performance, and to isolate any diversification gains. We also utilize a panel framework that allows measurement of the impact of changes in revenue diversification for a particular FHC. By moving to a "within" analysis, we avoid the omitted variable problem that is a concern in earlier, cross-sectional studies like DeYoung and Roland (2001) and Stiroh (2004b). This advantage, however, comes at the expense of introducing additional noise into the within analysis.

In terms of data, we examine the FHC, rather than subsidiary banks as in DeYoung and Roland (2001); Morgan and Samolyk (2003), and Stiroh (2004a,b). Even before GLBA, modern financial institutions operated with complicated structures that included tiered holding companies, commercial bank subsidiaries, and non-bank components like Section 20 subsidiaries.⁸ FHC managers, presumably, make strategic decisions with the performance of the entire operation in mind, and diversification benefits may exist for the institution as a whole, but not for individual subsidiaries. Moreover, the FHC is the relevant unit of observation from the perspective of a supervisor interested in systemic risk. Due to the "source of strength" doctrine, FHCs are responsible for their subsidiary banks, so regulators should be interested in the risk associated with the entire institution.⁹

Finally, we examine a relatively recent, short, and turbulent period from 1997:Q1 to 2002:Q4, which has several advantages. Earlier studies covered a period when

⁸ A Section 20 subsidiary was a bank holding company subsidiary that was engaged, to a limited degree, in prohibited activities like security underwriting and dealing.

⁹ The Federal Reserve explicates this doctrine in 12 CFR 225.4(a)(1) of Regulation Y and FRRS 4-878, a policy statement dated April 30, 1987.

banks operated in a more focused and regulated banking environment. The weak results of earlier studies could simply reflect an adjustment period when banks may not have had the scale or experience to reap the diversification gains, so looking at more recent data is useful. A short period is valuable because it alleviates some of the problems from consolidation, e.g., a given FHC becomes less comparable over time if it pursues acquisitions. In terms of turbulence, the Russian bond default and Long-Term Capital Management (LTCM)-induced liquidity crisis in 1998, the stock market boom and bust, and the slowdown in capital market activity led to increased uncertainty and several large FHCs explicitly stated that a more diversified strategy helped to stabilize revenue. If so, we should see more diversified FHCs outperforming their more concentrated peers during this period. Finally, we can compare and contrast these results to earlier studies that examined more stable periods, and thus build a more compelling case for the generality of the conclusions.

3. Definitions and data

The empirical analysis uses data on revenue sources and risk-adjusted performance for US bank and financial holding companies (FHCs). This section first defines the key variables in the analysis and provides some discussion of their relevance and importance. We then provide details on the construction of the dataset from the quarterly, FR Y-9C reports.

3.1. Variable definitions

3.1.1. Diversification measures

To measure the diversification of a FHC's activities, we follow the basic Herfindahl-type approach used in [Morgan and Samolyk \(2003\)](#); [Stiroh \(2004a\)](#) and [Thomas \(2002\)](#). Our primary measure of revenue diversification, DIV, accounts for variation in the breakdown of net operating revenue into two broad categories: net interest income, NET, and non-interest income, NON, which includes fiduciary income, fees and service charges, trading revenue, and other sources of non-interest income. Using this breakdown, we measure revenue diversification of the FHC as

$$\text{DIV} = 1 - (\text{SH}_{\text{NET}}^2 + \text{SH}_{\text{NON}}^2), \quad (1)$$

where SH_{NET} is the share of net operating revenue from net interest sources and SH_{NON} is the share of net operating revenue from non-interest sources defined as

$$\begin{aligned} \text{SH}_{\text{NET}} &= \frac{\text{NET}}{\text{NET} + \text{NON}}, \\ \text{SH}_{\text{NON}} &= \frac{\text{NON}}{\text{NET} + \text{NON}}. \end{aligned} \quad (2)$$

DIV measures the degree of diversification in a FHC's net operating revenue. A higher value indicates a more diversified mix: 0.0 means that all revenue comes from a single source (complete concentration), while 0.5 is an even split between net

interest income and non-interest income (complete diversification).¹⁰ These measures are then averaged over all quarters in a FHC's lifetime or over a specific year to get a measure of average revenue diversification, $\overline{DIV}$, average net interest income shares, $\overline{SH}_{NET}$, and average non-interest income shares, $\overline{SH}_{NON}$.

3.1.2. Risk-adjusted performance measures

Our primary measures of performance are based on profit ratios, e.g., return on equity, ROE, and return on assets, ROA, defined as annualized net income divided by equity and by total assets, respectively. For each FHC, we calculate the average ROE, $\overline{ROE}$, and average ROA, $\overline{ROA}$, over the quarters in a FHC's lifetime or a given year. For each FHC, we also calculate the standard deviation of ROE, σ_{ROE} , and the standard deviation of ROA, σ_{ROA} , over the same lifetime or year to measure the total volatility of profits.

These variables define risk-adjusted return on equity, RAR_{ROE} , and on assets, RAR_{ROA} , as

$$\begin{aligned} RAR_{ROE} &= \frac{\overline{ROE}}{\sigma_{ROE}}, \\ RAR_{ROA} &= \frac{\overline{ROA}}{\sigma_{ROA}}, \end{aligned} \quad (3)$$

where these ratios can be thought of as accounting returns per unit of risk.¹¹

Finally, we calculate a Z-score, which measures the number of standard deviations that profits must fall to drive a FHC into insolvency. The Z-score is essentially a measure of the distance to default for a given institution and is calculated as

$$Z\text{-score} = \frac{\overline{ROA + E/A}}{\sigma_{ROA}}, \quad (4)$$

where $\overline{E/A}$ is the average equity to assets ratio over the same period.

3.1.3. Discussion

It is useful at this point to discuss the appropriateness of these measures of risk and return. Finance theory suggests that firms should not utilize valuable resources to reduce risk because equity market investors can diversify and shed idiosyncratic (firm-specific) risk themselves by holding a well-diversified portfolio of stocks. This view, however, does not seem appropriate for FHCs due to financial market imperfections, agency costs, and informational asymmetries, so the total risk of an FHC, which is proxied by the unconditional volatility of profits described above, provides

¹⁰ This measure is analogous to a Herfindahl–Hirschman Index of concentration, except that the interpretation is reversed. Here, a larger number indicates a more diversified and less concentrated set of activities.

¹¹ These estimates are similar to a market-derived Sharpe Ratio, which defines risk-adjusted returns as market returns (less the risk-free rate) divided by the standard deviation of returns. Because our sample include all FHCs, we cannot create a measure of market returns. See Stiroh (2005) for that type of analysis.

relevant and useful information. Moreover, total risk and volatility are important concerns of other stakeholders including regulators/supervisors, management, and borrowers. We now briefly discuss the factors that make total risk a meaningful and relevant indicator of FHC performance.

From the regulator/supervisor perspective, there is a clear concern with the costs associated with failure, e.g., transaction and liquidation costs related to bankruptcy, systemic risk concerns, and costs to the insurance fund. As a result, supervisory interest is not in a market portfolio of diversified stocks, but rather in the total risk of the individual institutions that it supervises. This can be seen directly in Merton-type portfolio models of credit risk, developed by Merton (1974) and implemented in KMV risk models, which are driven by assumptions about total asset return volatility. Moreover, the bank's CAMELS rating include supervisors' assessment of a bank's ability to absorb future losses, the sensitivity of earnings to economics changes, and management's ability to measure, monitor, and control risk (Berger et al., 2001). Because the degree of supervisory oversight and regulatory burden depend on these ratings, managers will care about total risk through this supervisory channel.

From the perspective of shareholders, there is a long list of factors that make the textbook finance view less tenable and lead total risk to be relevant. Froot et al. (1993) and Froot and Stein (1998) highlight several of these channels. One, non-linear costs of external funds lead firm value to depend on the total volatility of returns, due to the possibility of foregone investment, and thus make active risk management good for shareholders. Two, certain risks cannot be easily traded or hedged, e.g., a firm may not be able to lay off the risk associated with the introduction of a new product, so this non-traded, idiosyncratic risk imposes real costs to owners. Third, costs of financial distress, e.g., bankruptcy costs, loss of value in asset sales, and search costs for new management, impose costs on owners and induce them to care about total risk. This is particularly true in banking where assets are relatively "opaque" and hard to value from the outside. Cebenoyan and Strahan (2004), for example, show that banks use loan sales to manage total risk and thus are able to operate in more risky lending markets. An additional factor is convexity in the corporate tax code, which increases firm value with cash-flow stability (Smith and Stulz, 1985). As a result, shareholders will care about total risk and the volatility of revenues.

From the manager's perspective, there are additional reasons that again make total risk a relevant metric. Stulz (1984) argues that risk-averse managers may have a large share of their wealth in the firm's stock, so that they care about the overall volatility and risk of the institution. Similarly, Cummins et al. (1998) point out that managers can't always diversify their skills or human capital, and thus manage risks and follow the total risk of the institution.

Finally, total risk is relevant from the point of view of the bank borrower due to the intrinsic value of bank relationships. Slovin et al. (1993) show that borrower stock prices fell after the *de facto* failure of Continental Illinois and interpret this as the evidence of the costs of severing intangible banking relationships that are valuable to borrowers. In this view, borrowers are bank stakeholders that care about the

total risk of the institution. Similarly, [Houston et al. \(1997\)](#) show that diversification of internal capital markets benefits borrowers through the efficient allocation of scarce capital resources. If banks have large variation in revenue, for example, positive net present value projects are not funded in periods of low realized cash flows. Bank-dependent borrowers, therefore, care about the volatility of revenues and total risk. This is particularly true for small firms that are dependent on small banks for financing ([DeYoung et al. \(2004\)](#)).

These arguments all suggest that it is worthwhile to examine the determinants of total risk of FHCs and indicate that the unconditional volatility of profits, as defined in Eqs. (3) and (4), are useful indicators of FHC performance. Of course, one could examine volatility by examining equity market returns. This has the advantage of being forward-looking, but comes at the expense of limiting the sample to a much smaller sample of the larger publicly-traded firms.¹² We use accounting data to include a larger sample and to be consistent and comparable with recent other work on the diversification and risk by [Cebenoyan and Strahan \(2004\)](#); [DeYoung and Roland \(2001\)](#); [Morgan and Samolyk \(2003\)](#), and [Stiroh \(2004a,b\)](#), and all of whom use similar measures of risk-adjusted returns.

3.2. Data

All data are taken from the holding company regulatory reports that are filed each quarter to the Federal Reserve, the FR Y-9C reports. These reports contain the complete balance sheet, income statement, and detailed supporting schedules for all US domestic bank holding companies and financial holding companies and, and thus provide information about the complete sources of revenue and financial performance of the institution. Due to the tiered nature of the US financial structure, i.e., one holding company may own another, we only include the top-tiered holding company and thus avoid multiple counting of the same activities.

[Table 1](#) presents summary statistics for our primary cross-sectional sample, i.e., one observation per US FHC where all variables are averaged over all quarters that we observe the FHC from 1997:Q1 to 2002:Q4. All data are deflated with the GDP deflator. We only include FHCs that meet the following criteria: at least eight quarters of data; all performance measures between the 1st and 99th percentile; and $0 \leq \overline{SH}_{\text{NON}} \leq 1$ (i.e., average non-interest income and average net interest income both positive). We use the regulatory FHC code as our primary identifier of each institution. When the FHC code changes, we treat this as a new organization because this change typically reflects a major structural reorganization and we do not want these types of changes to impact our measures of income volatility. This procedure left 1816 FHC observations, which accounted for 0.78 of the total assets and 0.74 of the FHCs in the original sample.

¹² This type of analysis has been done by [Demsetz and Strahan \(1997\)](#) and [Stiroh \(2005\)](#).

Table 1
Summary statistics for FHCs

	Mean	Median	Standard deviation	Minimum	Maximum
RAR_{ROE}	4.67	4.02	3.20	−0.19	18.43
RAR_{ROA}	4.64	4.02	3.19	−0.18	18.29
Z-score	42.59	36.60	27.02	2.93	149.50
Assets (\$m)	3896	321	33,000	38	930,000
Equity/assets	0.090	0.087	0.027	0.019	0.290
Loans/assets	0.641	0.655	0.116	0.040	0.937
ROA	0.011	0.011	0.004	−0.002	0.026
ROE	0.123	0.120	0.044	−0.040	0.269
SH_{NON}	0.199	0.178	0.103	0.014	0.873
DIV_{REV}	0.29	0.29	0.09	0.03	0.50
$SH_{fiduciary}$	0.075	0.090	0.129	−0.403	0.987
$SH_{service}$	0.468	0.465	0.233	−0.571	4.183
$SH_{trading}$	0.004	0.000	0.057	−0.687	1.933
SH_{other}	0.453	0.438	0.224	−3.183	1.958
$SH_{C\&I\ loan}$	0.177	0.160	0.109	0.000	0.974
$SH_{consumer\ loan}$	0.112	0.094	0.085	0.000	0.670
$SH_{real\ estate\ loan}$	0.651	0.666	0.162	0.000	1.000
$SH_{other\ loan}$	0.060	0.027	0.088	0.000	0.986

Results are for 1816 financial holding companies (FHCs) with at least eight quarters of data from 1997:Q1 to 2002:Q4. All variables are averages over the FHC's quarterly observations.

These FHCs were quite varied with mean average assets of \$3.9B and a range from \$38M to \$930B. As with the US banking system as a whole, this sample is dominated by a few large banks so the mean greatly exceeds the median (\$3.9B compared to \$0.32 billion). On the performance side, our sample includes both low- and high-performing FHCs. The mean $\overline{ROE}$ was 0.12, with a median of 0.12 and a range from −0.04 to 0.27, while the mean $\overline{ROA}$ was 0.011, with a median of 0.011 and range from −0.002 to 0.026. This variation will help to identify the link between operating strategies and performance.

A key component of our analysis is that we use the observed non-interest shares and diversification measure to identify differences in operating strategies between banks. Table 1 also shows the necessary variation across FHCs in both the average non-interest share ($\overline{SH_{NON}}$) and the average revenue diversification measure ($\overline{DIV}$). $\overline{SH_{NON}}$, for example, had a mean of 0.199, a median of 0.178, a standard deviation of 0.103 and a range from 0.014 to 0.873, while $\overline{DIV}$ had a mean of 0.29, a median of 0.29, a standard deviation of 0.09, and a range from 0.03 (nearly complete concentration) to 0.50 (equal revenue shares).

Our analysis identifies differences in operating strategies by differences in these measures across FHCs. A confounding factor, however, is that random fluctuations like unexpected macroeconomic shocks or industry fluctuations add noise to the FHC's income statement. To gauge the severity of this concern, we compared the variation across FHCs to the variation within FHCs. For both the non-interest share and the diversification measure, the variation across FHCs greatly exceeds the variation within individual FHCs. For example, the median standard deviation of

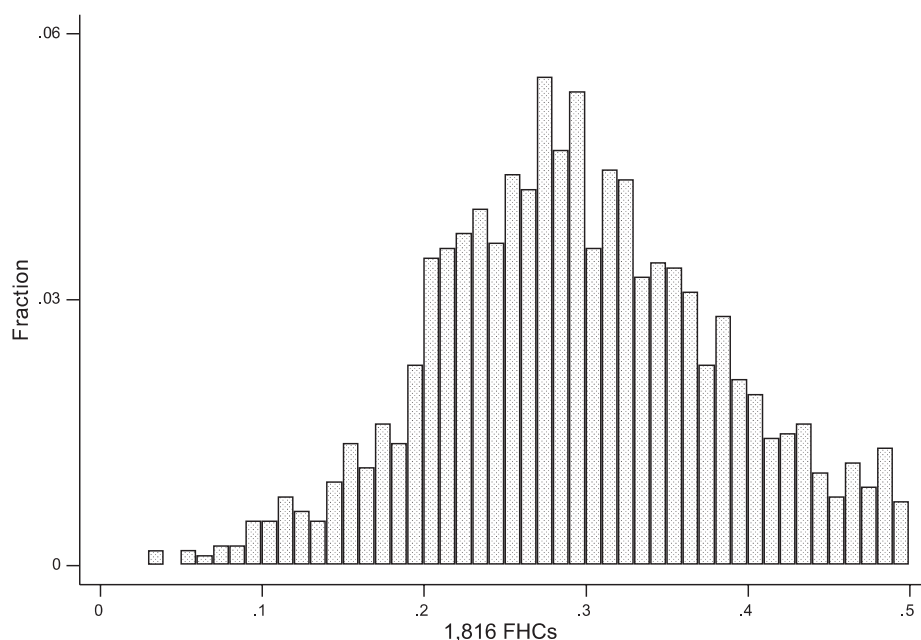


Fig. 1. Distribution of average revenue diversification.

non-interest income shares within FHCs, which measures the variation over a specific FHC's quarterly observations was only 0.031, compared to a standard deviation of 0.103 when one looks across FHCs.¹³ Similarly, the median standard deviation of revenue diversification within FHCs was 0.03 compared to a standard deviation of 0.09 across FHCs. We conclude that the observed non-interest shares and diversification measures in large part represent the operating choices of FHC managers, and not random fluctuations due to unexpected economic shocks.

To better gauge the diversification measure, Fig. 1 shows the distribution of average diversification for the main sample of 1816 FHCs. The distribution is roughly symmetric, but with a fatter right-hand tail.¹⁴ This figure shows that the majority of FHCs are somewhat diversified in terms of net operating revenue. Only 159 FHCs, for example, show an average diversification measure that is below 0.18, which corresponds to extreme concentration with a non-interest share of either less than 0.10 or greater than 0.90. Not surprisingly, the larger FHCs tend to be more diversified.

¹³ These estimates are constructed as follows. The standard deviation *across* FHCs (reported in Table 1) measures the variation in the distribution of average values of 1816 FHCs. Thus, there is one value for the entire sample. The standard deviation *within* an FHC measures time-variation over the observed quarters for each FHC. Thus, there is one value for each FHC in the sample, and we report the median of the 1816 observations.

¹⁴ The data strongly reject the null hypothesis of a normal distribution.

The concern about unexpected macroeconomic shocks can also be seen in the raw correlations between average profits and volatility of profits. While one might expect a positive correlation between profits and risk as bank managers earn higher returns for bearing risk, both the cross-section and annual estimates show a negative correlation between average profit ratios and the standard deviation of the ratio. This is true for both ROE and ROA, and is typically statistically significant. This suggests the presence of negative shocks that both lower average profits and increase volatility. These shocks, however, can emanate from non-interest income, net interest income, or non-interest expenses, so there is not an unambiguous link with the non-interest share and our main results should not be biased.

Finally, we construct variables that describe the focus within each broad activity class. For non-interest activities, we use income flows from the main non-interest activities – fiduciary income, fees and service charges, trading revenue, and other sources of non-interest income – to construct shares of total non-interest income, $SH_{\text{fiduciary}}$, SH_{service} , SH_{trading} , and SH_{other} , respectively. These shares show how focused a given FHC is on a particular non-interest activity. For net interest income, we are unable to isolate the corresponding income flows, so we turn to the balance sheet and use the loan amount for the main lending lines – commercial and industrial (C&I), consumer, real estate, and other – to construct shares of total loans, $SH_{\text{C\&I}}$, SH_{consumer} , $SH_{\text{real estate}}$, and SH_{other} , respectively. These shares show how focused a FHC is on a particular lending activity. As with the other variables, each share is averaged over all quarters in a FHC's lifetime and for the quarters within a year.¹⁵

4. Empirical framework and results

Our primary goal is to investigate the link between the diversification of a FHC's revenue and risk-adjusted performance. We do this in two ways. We first estimate between regressions where all variables are calculated over the FHC's lifetime and identification comes from differences *across* FHCs; this allows us to explicitly measure the volatility of profits over the FHC's lifetime. An advantage of this type of cross-sectional analysis is that random variation in income shares, which we use to identify diversification, are likely to average out and provide a relatively clear measure of an FHC's strategic choices. This approach, however, loses some information about how variables change over time and introduces a stronger possibility of an omitted variable bias. To address this, we also estimate panel regressions where all variables are calculated for the four quarterly observations in a given year. This gives us up to six observations for each FHC (one for each year) and provides an opportunity to estimate how risk-adjusted performance varies within FHCs due to changes in diversification and product line exposure. The downside, however, is that these measures are more likely to be noisy indicators of a bank's long-run strategic

¹⁵ Note that because the non-interest shares use income statement variables, they can be negative or greater than one. The summary statistics reported here include all observations, and the econometric results are robust to excluding these extreme values.

position. Nonetheless, this is a useful approach because it provides some new information about how performance varies with changes in revenue shares and focus.

4.1. Between analysis

The basic empirical specification for the cross-sectional, between regression is

$$Y_i = \alpha + \beta_1 \overline{\text{DIV}}_i + \beta_2 \overline{\text{SH}}_{\text{NON},i} + \gamma \overline{\text{X}}_i + \varepsilon_i, \quad (5)$$

where Y is a measure of performance, $\overline{\text{DIV}}$ is average revenue diversification, $\overline{\text{SH}}_{\text{NON}}$ is the average non-interest share, and $\overline{\text{X}}$ is a vector of other average control variables, all for FHC i . All variables are calculated over the FHC's lifetime so that we have one observation per FHC.

We are interested in the reduced-form relationship between revenue diversification and performance, but it is important to include the non-interest share directly because different activities are associated with very different ex post outcomes. For example, a FHC with 0.25 of its revenue from non-interest sources and a FHC with 0.75 would have the same $\overline{\text{DIV}}$ measure and would appear equally diversified, but these are very different operating strategies with different expected returns. Therefore, we include the non-interest share directly as an independent variable to control for this variation.

$\hat{\beta}_1$ and $\hat{\beta}_2$ can be interpreted as follows. $\hat{\beta}_1$ measures the impact of diversification where $\hat{\beta}_1 > 0$ if diversification is associated with improved risk-adjusted performance. $\hat{\beta}_2$ measures the effect of a shift away from net interest income and toward non-interest income where $\hat{\beta}_2 > 0$ indicates that marginal increases in non-interest income are associated with higher risk-adjusted performance.¹⁶

One issue is that $\overline{\text{DIV}}$ depends on $\overline{\text{SH}}_{\text{NON}}$ in a non-linear way. Given the definition of diversification in Eq. (1) and fact that $\text{SH}_{\text{NET}} + \text{SH}_{\text{NON}} = 1$ from Eq. (2), one can rewrite the definition of DIV as

$$\text{DIV} = 2\text{SH}_{\text{NON}} - 2\text{SH}_{\text{NON}}^2. \quad (6)$$

This specification with the diversification index and the non-interest share has several attractive features.¹⁷ First, it provides an intuitive way to decompose and interpret the impact of a change in strategy that alters the non-interest share. In particular, consider the impact of a change in $\overline{\text{SH}}_{\text{NON}}$ on Y :

$$\frac{\partial Y}{\partial \overline{\text{SH}}_{\text{NON}}} = \hat{\beta}_1 \frac{\partial \overline{\text{DIV}}}{\partial \overline{\text{SH}}_{\text{NON}}} + \hat{\beta}_2, \quad (7)$$

where the partial derivative is obtained from Eq. (6).

The first term on the right-hand side shows the indirect impact of a change in the non-interest share through changes in diversification. We refer to this as the *indirect*

¹⁶ Note that because the shares sum to one, one of the shares must be arbitrarily dropped. The coefficient on the included shares shows the impact of a 0.01 change from the omitted share to the included share.

¹⁷ While these variables are obviously correlated, this is not a debilitating econometric concern because of the non-linear relationship, i.e., econometricians routinely include a variable linearly and quadratically.

diversification effect of changes in non-interest income. Note that this effect depends on both the sign of $\hat{\beta}_1$ and the magnitude of the non-interest share: an increase in $\overline{SH}_{NON}$ will be diversifying ($\frac{\partial DIV_{REV}}{\partial \overline{SH}_{NON}} > 0$) if a FHC has an initial non-interest share below 0.50 and concentrating ($\frac{\partial DIV_{REV}}{\partial \overline{SH}_{NON}} < 0$) if a FHC has an initial non-interest share above 0.50. The second term on the right-hand side of Eq. (6) is the *direct exposure effect* of increased non-interest income shares. This measures the differences in ex post profits associated with the different activities. The sum of the indirect and direct effects is the *net effect* and shows how risk-adjusted performance changes with non-interest shares.

Second, this specification provides a clear parallel to a standard portfolio framework. If an investor can form a portfolio of two assets, A and B , then the expected return of the portfolio, $E(R_P)$, and variance of the portfolio, σ_P^2 , are:

$$\begin{aligned} E(R_P) &= wE(R_A) + (1 - w)E(R_B), \\ \sigma_P^2 &= w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 + 2w(1 - w)\text{Cov}(A, B), \end{aligned} \quad (8)$$

where $E(R)$ and σ^2 are the expected return and variances of the subscripted variables and $\text{Cov}(A, B)$ is the covariance between investment A and B .

If asset A offers a higher and more volatile return, then a shift toward A has several effects: higher expected portfolio returns because $E(R_A) > E(R_B)$, a direct increase in portfolio variance if the weighted variance of A exceeds the weighted variance of B , and an indirect diversification effect that depends on the shares and the sign of the covariance. In investment situations, one can measure the components of Eq. (8) directly and the question is typically to determine the optimal shares. In our application, we know the chosen shares, but data limitations prevent us from knowing the return on particular businesses. Therefore, we focus on the impact on these choices on bottom-line performance measures. Nonetheless, it is useful to keep this type of portfolio analysis in mind when thinking about the direct and indirect effects of changing focus.

A second issue is a potential bias in our variable construction. Net income is defined roughly as net interest income plus non-interest income less non-interest expense, while both the diversification measure and the non-interest share are functions of non-interest income and net interest income. This may cause bias, although the direction is ambiguous. *Ceteris paribus*, positive shocks to non-interest income would raise the non-interest share and raise profits (a positive bias on $\hat{\beta}_2$), while positive shocks to net interest income would lower the non-interest share and raise profits (a negative bias on $\hat{\beta}_2$). Stiroh (2004b) shows that non-interest income is the more volatile component and DeYoung and Roland (2001) present reasons for the higher volatility of non-interest income, e.g., switching costs and higher operating and financial leverage associated with these activities. Therefore, one might expect the positive bias to dominate.

As a robustness check, we also use the first period share as an indicator of the concentration and diversification of the FHC. This has the benefit of removing this

possible endogeneity bias, but a single period's value is likely to be a noisier signal of the FHC's strategic focus. That is, an average of non-interest shares over several quarters gives a better indication of whether the FHC is focused on lending activities or other activities that generate fees, trading revenue, and other income.

We also include a number of control variables in the $\bar{X}$ vector. These include the log of total assets, the equity to assets ratio, the loan to assets ratio, the growth of assets over the period (directly and as a quadratic), a dummy variable for the number of quarters the FHC is observed, dummy variables for each quarter in which a FHC operates, and a dummy variable for each state that is set equal to one if the FHC is headquartered in that state.

We include total assets to control for any systematic differences in performance across size classes, e.g., scale economies, geographic diversification, or different risk-management techniques. The equity ratio, loan ratio, and asset growth rates are included to control for other factors that are likely to affect performance, e.g., risk-loving banks may hold less equity, make more loans, and grow more rapidly, while loans may be more or less profitable than other earning assets. The number of quarter dummy controls for any survivor effect, e.g., long-lived FHCs may be more profitable. The quarter dummy and state fixed effect control for differences in the banking environment, although with multi-state FHCs the state dummy variables are a proxy at best.

Table 2 presents estimates of Eq. (5) using four performance measures (ROE , ROA , σ_{ROE} , and σ_{ROA}) as dependent variables. Using both measures of profits (columns 1 and 3), the data show no significant relationship between the average level of profits and either diversification or the non-interest share. In contrast, both variables are highly significant in the σ_{ROE} and σ_{ROA} regressions (columns 2 and 4). In both cases, the coefficient on $\overline{\text{DIV}}$ is negative as more diversified revenue portfolios are associated with less volatile profits. The coefficient on $\overline{\text{SH}}_{\text{NON}}$ is positive, indicating that an increased reliance on non-interest income is associated with more volatile profits.

Table 3 presents estimates using the risk-adjusted measures of financial performance (RAR_{ROE} , RAR_{ROA} , $Z\text{-score}$) as dependent variables. In all three cases, we find strong evidence that increased diversification improves risk-adjusted performance as the coefficient on $\overline{\text{DIV}}$ is positive and statistically significant. At the same time, however, the coefficient on $\overline{\text{SH}}_{\text{NON}}$ is negative and significant in all regressions, implying that an increased reliance on non-interest income is associated with reduced performance. Returning to the portfolio example, we interpret this as the negative impact from increased exposure to the more volatile asset (non-interest income activities) and the positive impact of diversification (through the covariance).

To find the net effect, one must consider both coefficients together because the typical FHC diversifies by increasing its non-interest share. Eq. (7) shows that the net effect of an increase of non-interest income equals the direct exposure effect ($\hat{\beta}_2$) plus the indirect diversification effect through changes in diversification ($\hat{\beta}_1 \frac{\partial \overline{\text{DIV}}}{\partial \overline{\text{SH}}_{\text{NON}}}$). Due to the non-linear relationship between $\overline{\text{DIV}}$ and $\overline{\text{SH}}_{\text{NON}}$, the magnitude of the net effect depends on the value of $\overline{\text{SH}}_{\text{NON}}$ used in evaluating Eq. (7).

Table 2
Mean and volatility of profitability regressions

	Return on equity (ROE)		Return on assets (ROA)	
	Mean	Std. dev.	Mean	Std. dev.
DIV _{REV}	−0.021 (0.034)	−0.093*** (0.027)	0.001 (0.003)	−0.007*** (0.002)
SH _{NON}	0.038 (0.032)	0.150*** (0.027)	0.002 (0.002)	0.012*** (0.002)
ln(Assets)	0.003*** (0.001)	−0.001 (0.001)	0.000*** (0.000)	0.000 (0.000)
Equity/assets	−0.337*** (0.039)	−0.361*** (0.046)	0.075*** (0.004)	0.009** (0.004)
Loans/assets	0.017* (0.010)	0.040*** (0.010)	0.002** (0.001)	0.003*** (0.001)
Asset growth	0.120 (0.094)	−0.244** (0.098)	0.022*** (0.006)	−0.019*** (0.007)
Asset growth ²	−1.401* (0.829)	2.047** (0.891)	−0.302*** (0.050)	0.169*** (0.059)
No. obs.	1816	1816	1816	1816
Adjusted R ²	0.170	0.171	0.318	0.117

OLS regressions include all FHCs with at least eight quarters of data, mean of all dependent variables within the 1st and 99th percentile, and average non-interest shares between 0.0 and 1.0. All regressions include state fixed effects, quarter fixed effects, and dummy variables for number of quarters the FHC is observed. All variables are average for all quarterly observations for each FHC. Robust standard errors are in parentheses.

***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Table 4 reports estimates of the direct, indirect, and the net effect from the RAR_{ROE} regression, evaluated at the 10th, 25th, 50th, 75th, and 90th percentile of SH_{NON} .¹⁸ The first row shows the direct effect: a 0.01 increase in non-interest income is associated with a significant decline of 0.06 in RAR_{ROE} . This does not vary across non-interest shares because the estimated relationship is linear. The second row shows the indirect effect, which varies monotonically from 0.06 (significant) for a FHC with a non-interest share at the 10th percentile to 0.03 (significant) for a FHC at the 90th percentile. This pattern across non-interest shares is reasonable: all of the shares are below 0.50, so an increase in non-interest income diversifies the revenue stream and improves performance. Moreover, the FHCs that are the most concentrated in net interest income (the 10th percentile) have the largest potential diversification gains from a shift toward non-interest income, while FHCs that have relatively more non-interest income (the 90th percentile) have fewer potential diversification gains.

¹⁸ The 10th, 25th, 50th, 75th, and 90th percentile correspond to average non-interest shares of 0.11, 0.14, 0.18, 0.23, and 0.30, respectively.

Table 3
Risk-adjusted performance regressions

	RAR _{ROE}	RAR _{ROA}	Z-score
DIV _{REV}	3.927** (1.557)	3.887** (1.571)	29.543** (11.779)
SH _{NON}	−6.156*** (1.305)	−6.600*** (1.311)	−61.181*** (9.134)
ln(Assets)	0.165** (0.084)	0.210** (0.085)	0.749 (0.625)
Equity/assets	16.119*** (2.952)	17.388*** (2.960)	307.974*** (28.216)
Loans/assets	−0.586 (0.753)	−1.004 (0.773)	−15.701** (6.344)
Asset growth	10.348 (6.390)	11.713* (6.159)	97.926*** (37.734)
Asset growth ²	−193.613*** (55.789)	−186.595*** (53.983)	−1433.838*** (282.656)
No. obs.	1816 0.102	1816 0.103	1816 0.189

OLS regressions include all FHCs with at least eight quarters of data, mean of all dependent variables within the 1st and 99th percentile, and average non-interest shares between 0.0 and 1.0. All regressions include state fixed effects, quarter fixed effects, and dummy variables for number of quarters the FHC is observed. All variables are average for all quarterly observations for each FHC. Robust standard errors are in parentheses.

***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

The final line shows the net effect. For a FHC with relatively low non-interest income, the positive gains from a more diversified revenue stream offset the negative effect from more non-interest income, although the net effect is not significantly different from zero. For FHCs that are more concentrated in non-interest income, however, the potential diversification gains are smaller, the negative effect dominates, and further shifts toward non-interest income are associated with declining performance. This net effect becomes significantly below zero between the 50th and 75th percentile.

These results show the double-edged nature of the trend toward non-interest income: increased revenue diversity does bring benefits, but there are offsetting effects from a greater reliance on the more volatile activities. This interpretation is similar to [Demsetz and Strahan \(1997\)](#), who argue that banks use up their diversification benefits by holding less equity capital making more high-risk commercial and industrial loans.

These results are estimated from a turbulent period in the late 1990s and early 2000s, but they are similar to results by [DeYoung and Roland \(2001\)](#) and [Stiroh \(2004b\)](#) who examined earlier periods. [DeYoung and Roland \(2001\)](#) attribute the increased volatility to switching costs, operating leverage, and financial leverage, all of which make non-interest income more volatile from 1988 to 1995, while [Stiroh \(2004b\)](#) shows that trading revenue, which is the most volatile part of non-interest

Table 4

Estimated impact of a change in the non-interest share on RAR_{ROE}

	Non-interest share percentiles				
	10th	25th	50th	75th	90th
Direct effect	−0.062*** (0.013)	−0.062*** (0.013)	−0.062*** (0.013)	−0.062*** (0.013)	−0.062*** (0.013)
Indirect effect	0.062** (0.025)	0.057** (0.023)	0.051** (0.020)	0.042** (0.017)	0.031** (0.012)
Net effect	0.000 (0.016)	−0.005 (0.015)	−0.011 (0.012)	−0.020* (0.010)	−0.031*** (0.008)

Estimates are based on regression results reported in Table 3, column 1 and evaluated at different values of the average non-interest share based on percentile ranks. The 10th, 25th, 50th, 75th, and 90th percentile correspond to average non-interest shares of 0.11, 0.14, 0.18, 0.23, and 0.30, respectively. Direct effect is estimated impact of a 1% increase in the non-interest income share. Indirect effect is estimated impact of a change in revenue diversification from a 1% increase in the non-interest income share. Net effect sums the direct and indirect effects. Robust standard errors are in parentheses.

***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

income, is an important source of the risk-adjusted profit from 1984 to 2001. The consensus across sample periods makes a compelling case that the trend toward non-interest activities has not improved risk-adjusted performance on net.

We next present robustness checks of the RAR_{ROE} regression that use alternative data definitions or sub-samples of the data.¹⁹ The goal here is to eliminate alternative explanations for the strong, negative correlation between risk-adjusted performance and non-interest shares by excluding the FHCs for which the alternative explanations are most likely. One concern, for example, is that this relationship could reflect a reverse causality story as poor-performing FHCs increase risk to try and recover profitability. We can address this concern by using the non-interest shares from the beginning of the observation period, by looking only at the sub-sample of successful and profitable FHCs, or by looking only at the sub-sample of long-lived FHCs.²⁰ A second concern is that acquisitions or fast internal growth may be associated with more volatile and worse performance. If these institutions also focus on non-interest income, then we would find the same conditional correlation. Large FHCs, for example, were active acquirers in the late 1990s and have relatively high non-interest shares, so merger-related adjustment costs could lead to both lower profits and increased volatility. We address this concern by looking only at “non-jumping” FHCs (asset changes of less than 0.20 in every quarter) over the sample period.²¹ Similarly, FHCs may be shifting their loan portfolios toward more risky

¹⁹ We only report estimates for the risk-adjusted return on equity, which is our preferred measure of performance. Results using the risk-adjusted return on assets and the Z-score are similar and available from the authors upon request.

²⁰ Another solution is to employ panel data methods, which is addressed in the following section.

²¹ Note there should not be any mechanical level effects from mergers because all variables are ratios.

loans at the same time that non-interest shares are rising, which would increase overall FHC risk. As a check, we examine a subset of FHCs that showed relatively stable shares over the period for the four major loan categories: consumer, C&I, real estate, and other. Finally, we examine whether the relationship holds for all FHCs, or whether the relationship differs for FHCs that are either very large or very small.

Table 5 presents results.²² Column 1 repeats the estimates for the full sample for ease of comparison. Column 2 uses the first-period values of all explanatory variables, Column 3 includes only profitable FHCs with $\overline{ROE} > 0$, and Column 4 includes only the FHCs with 24 quarters of data (the complete time period). In all cases, the estimated coefficients change very little, which makes it less likely that the observed relationships primarily reflect an increase in risk-taking by poor-performing FHCs. Moreover, the consistency of the results for the sub-sample that includes FHCs with 24 quarters of data suggests that noise in the diversification measure from volatile non-interest income is not driving the results. Column 5 includes only the non-jumping FHCs and the results remain robust; it does not appear that adjustment costs or acquisition-related effects are driving the results.²³ Column 6 includes only FHCs with relatively stable loan shares and shows similar results, so changes in the portfolio do not account for this relationship.²⁴

Columns 7 and 8 report results for small FHCs (average assets below the 25th percentile of the full sample) and large FHCs (average assets above the 75th percentile of the full sample), respectively. Here, we do see differences. Beginning with the small FHCs, the coefficients on both $\overline{DIV}$ and $\overline{SH}_{NON}$ are much larger (in absolute value) than for the full sample, but somewhat less precisely estimated. For the largest FHCs, the coefficients are smaller and not significantly different from zero.

These differences suggest that the small and medium-sized FHCs face the biggest gains from diversification and the biggest costs from increased exposure to non-interest income. One interpretation is that the larger FHCs, which have been involved in these activities for a longer period of time, have had time to reach the optimal level of diversification so marginal increases do not improve risk-adjusted performance. Moreover, they are more likely to have implemented the business practices and operations needed to be successful, and to be diversified geographically and across sectors. For the small and medium-size FHCs, there appears to be some unexploited opportunities to gain from diversification, although the direct costs associated with increased non-interest income are also apparent. Finally, we note that asset size is positively linked with performance for small FHCs, but not the large FHCs, which suggests some economies of scale at the low end of the size distribution.

²² Robustness results using the average profit ratios and the standard deviation of the profit ratios individually as the dependent variable yield similar results as in Table 2 and are available from the authors upon request.

²³ The primary result holds if we include only FHCs with average ROE above the 1st, 5th, 10th, 25th, or 50th percentile, rather than average ROE greater than zero or if we use a 0.05, 0.10, or 0.50 growth rate for the cut-off of our non-jumping sample, rather than the 0.20 cut-off reported.

²⁴ The same relationship holds if we use relative rankings of changes in loan shares, e.g., dropping FHCs with changes in shares below the 5th or above the 95th percentile.

Table 5
Robustness tests for RAR_{ROE} regressions

	Full sample	First-period values	Sub-samples					
			Profitable	Full-period	Non-jumping	Stable Loans	Small	Large
DIV_{REV}	3.927** (1.557)	3.400** (1.338)	3.784** (1.558)	4.233* (2.333)	4.441** (2.017)	5.643*** (1.929)	9.601 (6.366)	0.876 (3.595)
SH_{NON}	−6.156*** (1.305)	−5.391*** (1.074)	−5.885*** (1.304)	−5.543*** (1.828)	−6.433*** (1.917)	−8.480*** (1.746)	−14.046** (6.977)	−2.819 (2.390)
$\ln(\text{Assets})$	0.165** (0.084)	0.124 (0.086)	0.152* (0.083)	0.172 (0.113)	0.307** (0.125)	0.228** (0.110)	2.033** (0.803)	−0.093 (0.160)
Equity/assets	16.119*** (2.952)	8.098*** (2.685)	15.430*** (2.958)	17.713*** (4.755)	23.048*** (3.488)	15.044*** (3.882)	30.102*** (4.893)	10.460 (7.311)
Loans/assets	−0.586 (0.753)	−1.165* (0.708)	−0.379 (0.752)	0.121 (1.194)	−0.681 (0.845)	−0.774 (0.934)	−0.568 (1.340)	−0.852 (1.687)
Asset growth	10.348 (6.390)	7.056 (5.934)	9.145 (6.168)	57.909*** (15.622)	89.435*** (12.117)	30.497*** (10.274)	71.982*** (24.217)	3.249 (17.013)
Asset growth ²	−193.613*** (55.789)	−171.646*** (52.742)	−188.856*** (53.067)	−888.075*** (220.245)	−1213.412*** (173.086)	−411.077*** (93.695)	−1243.220** (522.949)	−171.233 (123.817)
No. obs.	1816	1814	1805	882	1,425	1,181	454	454
Adjusted R^2	0.102	0.090	0.101	0.071	0.128	0.109	0.210	0.028

OLS regressions include all FHCs with at least eight quarters of data, mean of all dependent variables within the 1st and 99th percentile, and average non-interest shares between 0.0 and 1.0. All regressions include state fixed effects, quarter fixed effects, and dummy variables for number of quarters the FHC is observed. All variables are average for all quarterly observations for each FHC, except column 2 which uses the first-period values of the explanatory variables. Profitable FHCs have average ROE greater than 0. Full-period FHCs have 24 quarters of data. Non-jumping FHCs have no observed growth rates greater than 20% or less than 20%. Stable loan FHCs have changes in loan shares over the period below 10 percentage points. Small FHCs have average assets below the 25th percentile (\$224 m) and large FHCs have average assets above the 75th percentile (\$641 m). Robust standard errors are in parentheses.

***. **. * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

We are also interested in how the impact of diversification may vary with other aspects of an FHC's characteristics and strategic choices. In particular, we are interested in how the impact of diversification may vary with size (large banks are more likely to have large off-balance sheet positions) and capital ratios (high charter value firms may have high capital ratios to protect that value).²⁵ Table 6 reports estimates of regressions with the diversification and non-interest share variable interacted with size (column 1), with the leverage ratio (column 2), and with both (column 3).

With the asset interaction terms, the results are insignificant and relatively uninformative due to the high collinearity between asset size, non-interest share, and the interaction term. With the capital ratio interaction, it appears that the benefits of diversification increase with the capital ratio. When both interactions are included, the capital ratios interactions are again significant, indicating that the benefits of diversification and non-interest exposure vary with leverage. These findings are consistent with the idea that high capital/high charter value FHCs have more incentive to reap diversification gains and protect their charter value.

Our final set of cross-sectional results more closely examines the underlying factors associated with the wide variation in performance. In these extended regressions, we include a detailed breakdown of lending and non-interest activities to sort out precisely which ones are more highly correlated with performance.²⁶ We calculate the share of the loan portfolio in major classes (real estate, commercial and industrial (C&I), consumer, and other loans) and the share of non-interest income from major categories (service charges, fiduciary, trading, fees and other income) and include these as additional explanatory variables in our risk-adjusted performance regressions.²⁷ The coefficients then tell us the predicted impact of a shift away from the omitted category to that particular activity.

Table 7 reports results for the three measures of risk-adjusted performance. The diversification and non-interest share coefficients remain the same sign and remain statistically significant (except for diversification in the Z-score regression). In terms of the loan shares, C&I loans and other loans are always negatively linked to risk-adjusted performance, while the coefficient on consumer loans changes signs and is imprecisely estimated. The low returns to C&I lending is also found in Cebenoyan and Strahan (2004) and Stiroh (2004a). In terms of the non-interest shares, fiduciary income is positive and highly significant, while trading and other income are negative. Trading is not quite significant (p -value of 0.10) in the RAR_{ROE} and Z-score regressions. The trading result is similar to Estrella (2001), who finds that mergers between banking and securities firms are less likely to produce gains because of

²⁵ We thank a referee for suggesting these additional interaction specifications and the interpretation of the results.

²⁶ Recall that we already include the loan share, which helps capture differences in the risk-adjusted returns on loans and other earning assets.

²⁷ These are the most detailed breakdown available in the regulatory Y-9C reports for FHCs. Note that because the shares sum precisely to 1.0, one share must be arbitrarily dropped. We dropped the largest share of loans (real estate) and in non-interest income (service charges). Note also that data limitations force us to use stock shares for loans and flow shares for non-interest activities.

Table 6
Interaction regressions

	Base case	Size interaction	Equity/assets interactions	Both interactions
DIV _{REV}	3.927** (1.557)	12.070 (11.412)	−9.248* (4.744)	−10.518 (13.093)
SH _{NON}	−6.156*** (1.305)	−2.928 (7.195)	−1.766 (3.667)	7.245 (9.113)
DIV _{REV} * ln(Assets)	−	−0.662 (0.848)	− (0.855)	−0.045
SH _{NON} * ln(Assets)	−	−0.205 (0.500)	− (0.522)	−0.521
DIV _{REV} * Equity/assets	−	− (46.209)	143.206*** (46.594)	152.596***
SH _{NON} * Equity/assets	−	− (32.185)	−46.074 (32.116)	−58.435*
ln(Assets)	0.165** (0.084)	0.490* (0.280)	0.179** (0.084)	0.363 (0.280)
Equity/assets	16.119*** (2.952)	16.166*** (2.962)	−12.915 (8.630)	−13.346 (8.722)
Loans/assets	−0.586 (0.753)	−0.670 (0.767)	−0.532 (0.741)	−0.642 (0.752)
Asset growth	10.348 (6.390)	9.717 (6.343)	13.085** (6.096)	12.330** (6.086)
Asset growth ²	−193.613*** (55.789)	−197.234*** (54.254)	−221.903*** (52.730)	−223.291*** (51.774)
No. obs.	1816	1816	1816	1816
Adjusted R ²	0.102	0.102	0.107	0.107

OLS regressions include all FHCs with at least eight quarters of data, mean of all dependent variables within the 1st and 99th percentile, and average non-interest shares between 0.0 and 1.0. RAR^{ROE} is the dependent variable in all regressions. All regressions include state fixed effects, quarter fixed effects, and dummy variables for number of quarters the FHC is observed. All variables are average for all quarterly observations for each FHC. Robust standard errors are in parentheses.

***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

the securities firms' highly volatile returns. These extended results show significant variation in the outcomes associated with different lending and non-interest strategies, but they don't change the main conclusion that diversification has some benefits that are offset by the costs of increased exposure to non-interest income activities.

4.2. Within analysis

The previous results provide evidence of diversification benefits across FHCs, but also of a strong negative correlation between a FHC's focus on non-interest income and its risk-adjusted performance. One generic concern with this type of

Table 7

Extended risk-adjusted performance regressions

	RAR _{ROE}	RAR _{ROA}	Z-score
DIV _{REV}	2.694* (1.500)	2.850* (1.513)	17.839 (11.558)
SH _{NON}	−5.130*** (1.195)	−5.663*** (1.167)	−54.278*** (8.521)
ln(Assets)	0.218*** (0.084)	0.255*** (0.086)	1.089* (0.637)
Equity/assets	14.682*** (2.902)	16.073*** (2.929)	295.195*** (27.923)
Loans/assets	−0.329 (0.766)	−0.696 (0.783)	−13.886** (6.450)
Asset growth	9.704 (6.699)	11.135* (6.398)	94.404** (39.265)
Asset growth ²	−173.095*** (59.041)	−167.072*** (56.339)	−1277.707*** (303.033)
SH _{fiduciary}	2.729*** (0.693)	2.572*** (0.715)	21.983*** (6.101)
SH _{trading}	−1.555 (1.302)	−1.782 (1.094)	−13.421 (8.290)
SH _{other}	−1.048** (0.461)	−1.112** (0.462)	−5.832 (4.465)
SH _{C&I loan}	−2.403*** (0.700)	−1.923** (0.775)	−16.479*** (6.271)
SH _{consumer loan}	−0.808 (0.974)	−0.892 (0.970)	2.098 (7.703)
SH _{other loan}	−2.374*** (0.901)	−1.871** (0.923)	−15.595** (7.844)
No. obs.	1816	1816	1816
Adjusted R ²	0.125	0.123	0.206

OLS regressions include all FHCs with at least eight quarters of data, mean of all dependent variables within the 1st and 99th percentile, and average non-interest shares between 0.0 and 1.0. All regressions include state fixed effects, quarter fixed effects, and dummy variables for number of quarters the FHC is observed. All variables are average for all quarterly observations for each FHC. Robust standard errors are in parentheses.

***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

cross-sectional result is that some omitted variable, like management ability, industry exposure, or geographic location, may be driving the results. As discussed by Campa and Kedia (2002), diversification is endogenous and thus simple cross-sectional estimates may be biased. For example, good managers may both diversify and earn high profits, but increases in diversification or non-interest income for a given FHC may not affect outcomes. Because FHC's change their operating strategies and their exposure to different business lines, we can explore this issue by exam-

ining how variation in strategic focus over a FHC's life affects risk-adjusted performance. An empirical difficulty, however, is that this short-term variation is more likely to be affected by unexpected economic shocks, e.g., the Russian bond defaults in 1998 and the recession in 2001, which might weaken the link between expected and actual returns.

We use the same underlying data as in the previous analysis, but we now treat each year of data for each FHC as a separate observation to create observations cross-classified by FHC and year. More precisely, we calculate averages and standard deviations over the quarters in each year rather than over the FHC's full lifetime and construct a panel of FHC/year observations. This allows us to include a fixed effect to capture unobserved heterogeneity, but comes at the expense of increased noise when mean and volatility are calculated over just four quarters of data.

The basic fixed effect regression is

$$Y_{i,t} = \alpha_i + \beta_1 \overline{\text{DIV}}_{i,t} + \beta_2 \overline{\text{SH}}_{\text{NON},i,t} + \gamma X_{i,t} + \varepsilon_{i,t}, \quad (9)$$

where $Y_{i,t}$ is some performance measure, α_i is a FHC fixed effect, and variables are means of the quarterly observation in year t for FHC i .

Our sample includes 8770 observations for 2226 distinct FHCs that meet the following criteria: all dependent variables between the 1st and 99th percentile of all observations; $0 \leq \overline{\text{SH}}_{\text{NON},i,t} \leq 1$; and four quarters of data in each annual observation. The average number of observations for each FHC is just under 4, the mean asset size of the observations was nearly \$4B, and the mean annual ROE was 0.126.

Table 8 presents results for the three primary measures of risk-adjusted performance (RAR_{ROE} , RAR_{ROA} , Z -score). In all cases, the coefficient on revenue diversification is far from statistical significance, indicating no impact from changes in diversification within individual FHCs. In contrast, the average non-interest share remains negative and significant in all three regressions: when a FHC has above-average non-interest income, its risk-adjusted performance is below average.

One interpretation of the divergence in the between and within estimates (Table 3 vs. Table 8) is that the large differences across FHCs matter most for performance, but that these reflect omitted variables other than are correlated with diversification. Campa and Kedia (2002), for example, found that the diversification discount disappears when one controls for the endogenous decision to diversify and attribute this to omitted firm characteristics that drive both the diversification decision and firm performance. In this view, diversification itself does not drive performance. Schoar (2002) also reports a much smaller within effect in terms of productivity (i.e., diversified firms are more productive, but increases in diversification lower productivity), but offers an alternative interpretation. She attributes this to a "new toy effect" as management shifts focus toward newly-acquired segments. In this view, there is a real negative impact from diversification due to changes in management behavior.

The results presented here are similar to both studies and clearly show that the impact of diversification is largest across FHCs, but it is not clear whether this reflects changes in management focus or an omitted variable. In sharp contrast, the strength and stability of the non-interest share coefficient in the between and within analysis confirms the robust, negative link between a focus on non-interest income

Table 8
Fixed effect regressions

	RAR_{ROE}	RAR_{ROA}	Z-score
DIV_{REV}	−2.092 (4.839)	0.046 (4.593)	5.307 (39.098)
SH_{NON}	−8.962* (4.609)	−9.703** (4.374)	−98.408*** (37.240)
$\ln(\text{Assets})$	−1.350* (0.795)	−1.844** (0.754)	−11.160* (6.421)
Equity/assets	0.046 (11.468)	−5.776 (10.885)	297.886*** (92.668)
Loans/assets	5.996** (2.597)	5.472** (2.465)	5.553 (20.981)
Asset growth	−14.008*** (4.726)	−13.761*** (4.486)	−78.737** (38.189)
Asset growth ²	−120.460*** (26.624)	−131.652*** (25.269)	−1241.437*** (215.124)
No. obs.	8772	8772	8772
Within- R^2	0.023	0.028	0.025

Regressions include all FHC observations with four quarters of data in a year, mean of all dependent variables within the 1st and 99th percentile, and annual non-interest shares between 0.0 and 1.0. All variables are average for all quarterly observations for each FHC in a given year. Robust standard errors are in parentheses.

***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

and performance. Increases in non-interest income seem to unambiguously lower risk-adjusted profits as there is a negative direct effect and no positive diversification effect.

An alternative interpretation is more mechanical and reflects a divergence between expected and actual profits. That is, diversified FHCs take on additional risk because they expect higher returns, but these are not always realized due to exogenous shocks to economic conditions.²⁸ The cross-sectional results in Table 3 essentially average performance over a longer time period (6 years) and show the positive gains of diversification as expected returns are more closely aligned with realized returns. In contrast, the panel results in Table 8, which average over the four quarters in a given year, are more susceptible to unexpected shocks to performance that may drive actual returns below expected returns.

Table 9 compares alternative specifications and sub-samples to better understand these relationships.²⁹ The first column is a simple OLS regression (dropping the FHC

²⁸ Demsetz and Strahan (1997) show that bank holding companies use diversification gains to take on more risk in the form of lower capital ratios and higher risky loan shares.

²⁹ We only report estimates for risk-adjusted return on equity. Results using the risk-adjusted return on assets and the Z-score are similar and available from the authors upon request.

Table 9
Alternative specifications for RAR_{ROE} regressions

	OLS	Fixed effects	Between effect	Fixed effect sub-samples		
				Full-period	Profitable	Non-jumping
DIV_{REV}	6.576*** (2.233)	−2.092 (4.839)	7.366** (2.998)	−4.989 (6.993)	−1.946 (4.876)	−2.042 (4.837)
SH_{NON}	−11.359*** (2.069)	−8.962* (4.609)	−13.718*** (2.594)	−6.750 (6.391)	−9.025* (4.649)	−8.369* (4.609)
$\ln(\text{Assets})$	1.933*** (0.130)	−1.350* (0.795)	1.994*** (0.145)	−1.755* (1.066)	−1.425* (0.803)	−1.600** (0.814)
Equity/assets	28.093*** (3.591)	0.046 (11.468)	31.059*** (5.766)	−13.705 (16.690)	−4.390 (11.704)	−2.631 (11.754)
Loans/assets	1.920* (0.982)	5.996** (2.597)	1.437 (1.412)	0.720 (3.514)	6.115** (2.619)	5.891** (2.599)
Asset growth	−2.514 (8.416)	−14.008*** (4.726)	12.460 (7.704)	−0.693 (8.911)	−6.802 (5.901)	10.703 (6.675)
Asset growth ²	−182.670*** (55.209)	−120.460*** (26.624)	−244.332*** (40.929)	−225.680*** (58.448)	−170.335*** (35.361)	−399.297*** (56.613)
No. obs.	8772	8772	8772	4260	8720	8741
R^2	0.112	0.023	0.206	0.020	0.024	0.026

Regressions use RAR_{ROE} as the dependent variable and include all FHC observations with four quarters of data in a year, mean of all dependent variables within the 1st and 99th percentile, and annual non-interest shares between 0.0 and 1.0. All variables are average for all quarterly observations for each FHC in a given year. OLS reports R^2 . Fixed effects regression includes a FHC-specific fixed effect and reports within- R^2 . Between effects regressions averages all annual observations for each FHC and reports between R^2 . Sub-sample regressions include a FHC-specific fixed effect and the following sets of FHCs: full period sub-sample includes only FHCs with 24 quarters of data; profitable sub-sample includes only FHCs with annual profits greater than zero; and non-jumping sample includes only FHCs with annual growth between −20% and 20%. Robust standard errors are in parentheses.

***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

fixed effect) with all 8772 panel observations. The main results from the earlier cross-sectional analysis re-emerge: risk-adjusted profits are positively related to revenue diversification, but negatively related to the non-interest share. When we include an FHC fixed effect (column 2, and in Table 8), the positive diversification effect vanishes but the negative non-interest share effect remains. Column 3 reports the between regression (suppresses the time dimension and averages annual observations across FHCs), which yields similar conclusions as in the pure cross-section regression in Table 3. The change in estimated coefficients between the OLS, fixed effect, and between regressions shows that apparent diversification benefits are primarily found looking across FHCs, rather than within FHCs over time.

The remaining columns provide further robustness checks by limiting the analysis to the similar sub-sets of data as above. Because we are primarily interested in how changes in strategy by a given FHC affect performance, we only report fixed effect

results. For long-lived FHCs (column 4), for profitable FHCs (column 5), and non-jumping FHCs (column 6), the same results hold: no obvious diversification gains, but a strong negative link with the non-interest income share.

These panel estimates cast doubt on the notion that continued shifts toward non-interest income will improve performance. Within FHCs – the dimension that individual FHC managers are most interested in – we find no link between changes in diversification and changes in performance, but a large negative link between performance and the non-interest share. The strength and robustness of the relationship suggests that the high volatility of certain non-interest activities like fees and trading makes these activities less profitable (on a risk-adjusted basis) than interest-generating activities.

5. Conclusions

This paper examines the link between revenue diversification and risk-adjusted performance. Our main conclusion is that diversification benefits between FHCs are more than offset by increased exposure to non-interest activities, which are quite volatile but not more profitable than lending activities. Within FHCs, marginal increases in revenue diversification are not associated with performance, which may reflect either a change in managerial focus or simply reflect the endogenous nature of the diversification decision. In contrast, marginal increases in non-interest income are still associated with declines in risk-adjusted profits, suggesting a very robust relationship.

This is the darkside of diversification as FHCs move into activities that are inherently more volatile, which has implications for a range of FHC stakeholders. Regulators, for example, should be concerned with more volatile revenue streams that affect default probability, while shareholders should be aware of the potential implications for firm value due to the potential for missed investment opportunities and increased costs. Similarly, managers and borrowers could be hurt by wide swings in realized revenue due to inefficient or restricted allocation of credit. This is likely to be particularly true for small firms that are dependent on small banks.

This conclusion begs the question of why are FHCs moving into these activities. One might initially think that normal competitive forces have simply eliminated profits. While this explains the lack of excess returns for diversified FHCs, it is hard to reconcile with the finding that more diversified FHCs have consistently higher volatility and lower risk-adjusted profits and suggests that other factors are at work. One explanation is that FHCs may have simply gotten the diversification idea wrong. Many FHCs, for example, have pointed to “cross-selling” as a key strategic means to lower costs, increase income, and diversify revenue. If FHCs are really trying to diversify revenue by selling many products to the same customers, then this may simply expose multiple businesses to the same shocks, increase the correlation across revenue streams, and reduce potential diversification benefits. Moreover, FHCs are shifting into precisely those activities that are the most volatile, which offsets any diversification benefits.

An alternative interpretation is that FHC managers are more interested in expected returns than in the volatility. If FHC managers are large equity-holders, for example, they might take risk beyond what debt-holders and supervisors would prefer. This could be exacerbated by any implicit government guarantee (e.g., a firm considered “too-big-to-fail”), that reduces the incentives for debt-holders to monitor and discipline managers. This is especially true for the largest FHCs, which have in fact shifted the most into the highly volatile activities. Another reason for excess risk-taking is the standard principal-agent explanation: traders, brokers, and underwriters (agents) like volatility more than FHC shareholders (principals) do.

Short-run phenomenon may have also contributed to the unprofitable shift toward non-interest income. For example, FHC managers may have over-reacted to the lending problems of the late 1980s and early 1990s by shifting too far toward new activities. This desire to avoid earlier problem areas like real estate lending, coupled with financial innovation and deregulation that opened new markets and products, may have led the FHCs to push too far into these new activities. Other motives like empire-building and managerial misconduct would also contribute as managers grow their operations and expand their control of new businesses beyond profit-maximizing levels. Alternatively, these activities may ultimately be profitable, but adjustment costs could hold down the short-run returns. For example, FHCs may need time to build the business practices, scale, technology, and expertise to successfully combine these different products and achieve higher risk-adjusted profits. In this view, improved performance for the diversified FHC will emerge with some delay.

A final explanation is that the period examined (1997–2002) could have been just a bad draw for the FHCs. The combination of stock market crash, telecom bust, slowdown in initial public offerings, and Russian debt crisis may have combined to make this a period of particularly low returns for non-interest activities. DeYoung and Roland (2001) and Stiroh (2004b), however, have reported similar results for earlier periods, so this explanation appears less plausible.

At this stage it is difficult to sort out the remaining explanations, although anecdotal evidence suggests that FHC managers are realizing that a diversified firm will not guarantee success. Some FHCs, for example, have recently indicated a strategy to shift away from acquiring additional business lines, which was a major focus in the late 1990s, and toward focusing on how to derive greater profits from businesses lines they already own and operate. Moreover, several large FHCs have recently retrenched and exited from businesses that they have just entered, e.g., US Bancorp and FleetBoston each shed an investment bank subsidiary and Citigroup spun off an insurance arm.³⁰ There is also some preliminary evidence that this is a developing trend for FHCs: the number of announced bank and thrift acquisitions of financial service companies rose from 1997 to 2000 and has since dramatically fallen off.³¹

³⁰ A Painless Extraction: Closer Look at USB-Piper, Laura Mandaro, *American Banker*, February 21, 2003.

³¹ Financial service companies include finance companies, mortgage banks, brokers/dealers and investment advisors. Data comes from the annual supplement to SNL Bank Mergers and Acquisitions Scoreboard (Winter 1997–2002).

This is certainly consistent with the claim that FHC manager care about total risk, but whether this is part of a bigger trend or just reflects changing economic conditions and opportunities is an interesting question for future work.

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